

begins. Ideally the team has been involved both in the identification of the customer needs and in the setting of the target product specifications. Those members of the team who were not involved in these preceding steps should become familiar with the processes used and their results before concept generation activities begin. (See Chapter 5, Identifying Customer Needs, and Chapter 6, Product Specifications.)

As stated before, the challenge was to "design a better handheld roofing nailer." The scope of the design problem could have been defined more generally (e.g., "fasten roofing materials") or more specifically (e.g., "improve the speed of the existing pneumatic tool concept"). Some of the assumptions in the team's mission statement were:

- The nailer will use nails (as opposed to adhesives, screws, etc.).
- The nailer will be compatible with nail magazines on existing tools.
- The nailer will nail through roofing shingles into wood.
- The nailer will be handheld.

Based on the assumptions, the team had identified the customer needs for a handheld nailer. These included:

- The nailer inserts nails in rapid succession.
- The nailer is lightweight.
- The nailer has no noticeable nailing delay after tripping the tool.

The team gathered supplemental information to clarify and quantify the needs, such as the approximate energy and speed of the nailing. These basic needs were subsequently translated into target product specifications. The target specifications included the following:

- Nail lengths from 25 millimeters to 38 millimeters.
- Maximum nailing energy of 40 joules per nail.
- Nailing forces of up to 2,000 newtons.
- Peak nailing rate of one nail per second.
- Average nailing rate of 12 nails per minute.
- Tool mass less than 4 kilograms.
- Maximum trigger delay of 0.25 second.

Decompose a Complex Problem into Simpler Subproblems

Many design challenges are too complex to solve as a single problem and can be usefully divided into several simpler subproblems. For example, the design of a complex product like a document copier can be thought of as a collection of more focused design problems, including, for example, the design of a document handler, the design of a paper feeder, the design of a printing device, and the design of an image capture device. In some cases, however, the design problem cannot readily be divided into subproblems. For example, the problem of designing a paper clip may be hard to divide into subproblems. As a general rule, we feel that teams should attempt to decompose design problems, but should be aware that such a decomposition may not be very useful for products with extremely simple functions.

Dividing a problem into simpler subproblems is called *problem decomposition*. There are many schemes by which a problem can be decomposed. Here we demonstrate a *functional decomposition* and also list several other approaches that are frequently useful.

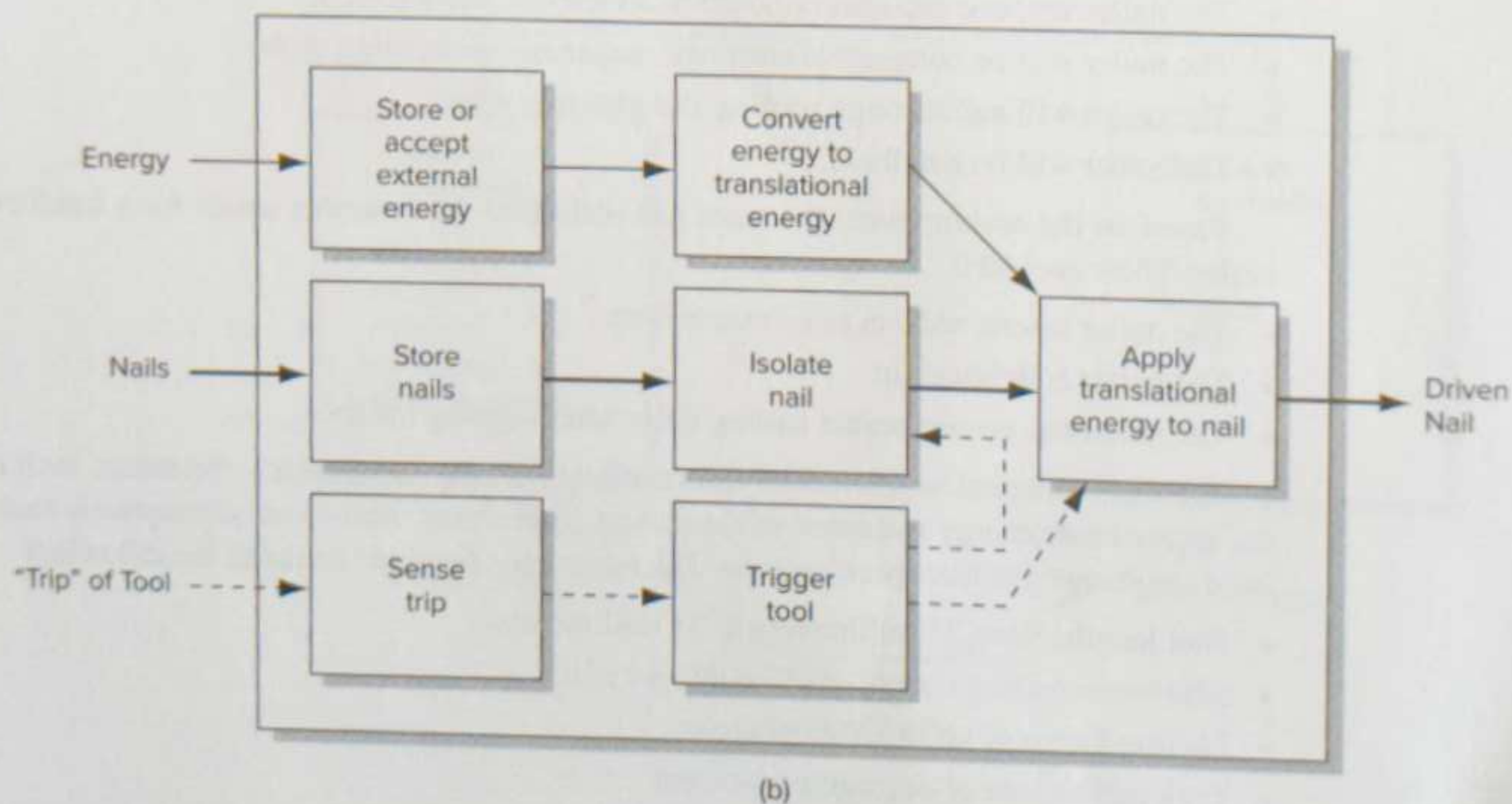
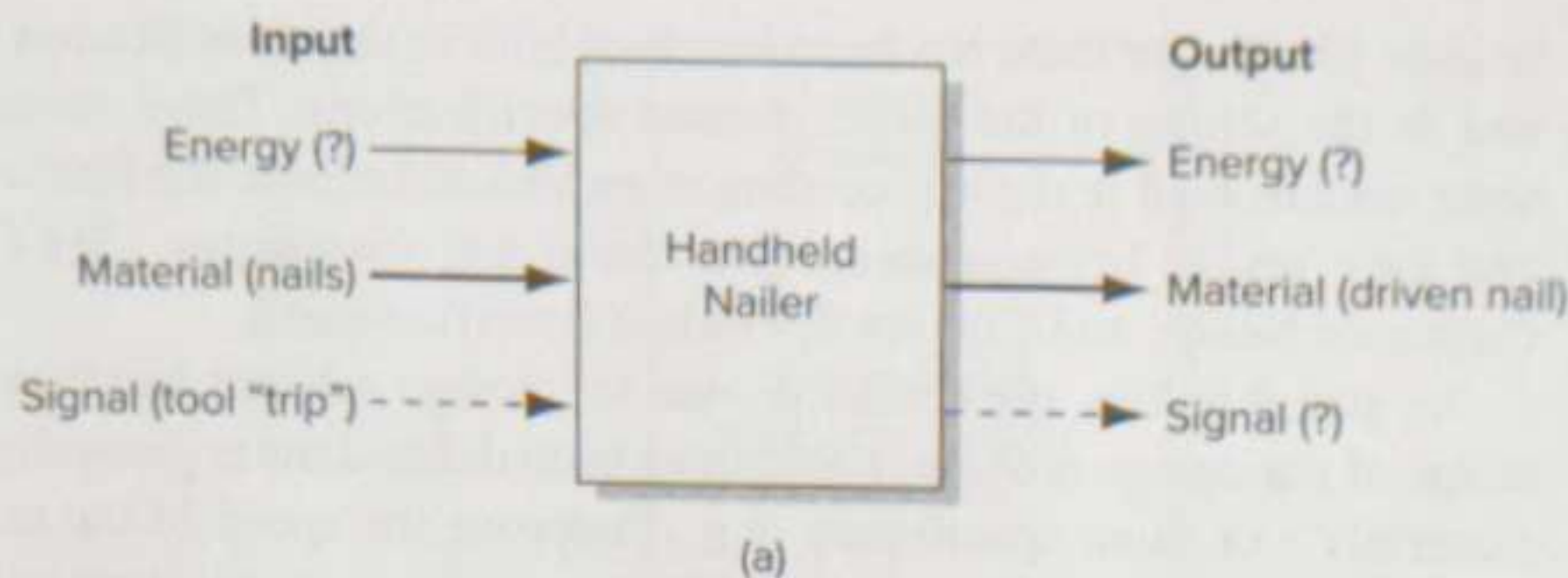


EXHIBIT 7-4 Function diagram of a handheld nailer arising from a functional decomposition: (a) overall “black box”; (b) refinement showing subfunctions.

The first step in decomposing a problem functionally is to represent it as a single *black box* operating on material, energy, and signal flows, as shown in Exhibit 7-4(a). Thin solid lines denote the transfer and conversion of energy, thick solid lines signify the movement of material within the system, and dashed lines represent the flows of control and feedback signals within the system. This black box represents the overall function of the product.

The next step in functional decomposition is to divide the single black box into subfunctions to create a more specific description of what the elements of the product might do to implement the overall function of the product. Each subfunction can generally be further divided into even simpler subfunctions. The division process is repeated until the team members agree that each subfunction is simple enough to work with. A good rule of thumb is to create between 3 and 10 subfunctions in the diagram. The end result, shown in Exhibit 7-4(b), is a function diagram containing subfunctions connected by energy, material, and signal flows.

Note that at this stage the goal is to describe the functional elements of the product without implying a specific technological working principle for the product concept. For example, Exhibit 7-4(b) includes the subfunction "isolate nail." This subfunction is expressed in such a way that it does not imply any particular physical solution concept, such as indexing the coil of nails into a slot or breaking a nail sideways off of the stick. The team should consider each subfunction in turn and ask whether it is expressed in a way that does not imply a particular physical solution principle.

There is no single correct way of creating a function diagram and no single correct functional decomposition of a product. A helpful way to create the diagram is to quickly create several drafts and then work to refine them into a single diagram that the team is comfortable with. Some useful techniques for getting started are:

- Create a function diagram of an existing product.
- Create a function diagram based on an arbitrary product concept already generated by the team or based on a known subfunction technology. Be sure to generalize the diagram to the appropriate level of abstraction.
- Follow one of the flows (e.g., material) and determine what operations are required. The details of the other flows can be derived by thinking about their connections to the initial flow.

Note that the function diagram is typically not unique. In particular, subfunctions can often be ordered in different ways to produce different function diagrams. Also note that in some applications the material, energy, and signal flows are difficult to identify. In these cases, a simple list of the subfunctions of the product, without connections between them, is often sufficient.

Functional decomposition is most applicable to technical products, but it can also be applied to simple and apparently nontechnical products. For example, an ice cream scoop has material flow of ice cream being separated, formed, transported, and deposited. These subfunctions could form the basis of a problem decomposition.

Functional decomposition is only one of several possible ways to divide a problem into simpler subproblems. Two other approaches are:

- **Decomposition by sequence of user actions:** For example, the nailer problem might be broken down into three user actions: moving the tool to the gross nailing position, positioning the tool precisely, and triggering the tool. This approach is often useful for products with very simple technical functions involving a lot of user interaction.
- **Decomposition by key customer needs:** For the nailer, this decomposition might include the following subproblems: fires nails in rapid succession, is lightweight, and has a large nail capacity. This approach is often useful for products in which form, and not working principles or technology, is the primary problem. Examples of such products include toothbrushes (assuming the basic brush concept is retained) and storage containers.

Focus Initial Efforts on the Critical Subproblems

The goal of all of these decomposition techniques is to divide a complex problem into simpler problems such that these simpler problems can be tackled in a focused way. Once problem decomposition is complete, the team chooses the subproblems that are most critical to the success of the product and that are most likely to benefit from novel or creative